Effort by M Arif Khan



BIO301 Current Final term Paper's questions solved by M Arif Khan. (2021)

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Short Question`s Answer

1. **What are the uses of PCR?**

Answer: The polymerase chain reaction has been elaborated in many ways. PCR is commonly used in following applications;

- Genotyping
- Cloning
- Mutation detection
- Sequencing
- Microarrays
- Forensics and paternity testing.

2. What are Multifactorial disease?

Answer:

Multifactorial diseases are caused by a combination of the effects of multiple genes or by interactions between genes and the environment. Genetic disorders controlled by many genes plus the effects of the environment.

- Asthma
- Cancer
- Diabetes
- Heart diseases
- Hypertension
- Inflammatory bowel disease
- Intellectual disability
- Obesity
- Infertility

3. Difference b/w DNA & cDNA

Answer: The key difference between DNA and cDNA is that DNA contains both exons and introns while the cDNA contains only exons.

4. **Function of STR?**

Answer

- Short Tandem repeats are often enriched in genes controlling body morphology.
- STRs have potentials to accelerate the evolution of gene expression and generate sufficient variability.
- STR describes a pattern that helps to determine inherited traits of individual.
- STRs can be very useful in determining parentage.
- STRs are used for certain genealogical DNA tests.

5. **How structure change in chromosomes?**

Answer

Changes in chromosome structure can come due to deletions, duplications and rearrangements. Loss of a segment of the chromosome is known as deletion, whereas

addition of a segment of chromosome on the same or different chromosome is known as duplication.

Extensive Questions

1. Explain Level of mutation. (5-marks)

Answer:

There are two basic levels of mutations.

- Two major levels of mutations are germ-line mutation and somatic mutation.
- Germ-line mutations occur in gametes, **while** Somatic mutations occur in other cells of the body.

Types of Mutations

There are three types of DNA Mutations;

1. ***Point Mutation. (Base Substitutions)***

   Single base substitutions are called point mutations which causes sickle-cell disease. Point mutations are two types.

   - **Transition:** If purines is replaced by purines and pyrimidines is replaced by pyrimidines.
   - **Transversion:** When a purine is substituted for a pyrimidine or a pyrimidine replaces a purine

2. ***Deletions***

   The type of mutation where one or more DNA base pairs are removed from a DNA sequence during DNA replication is called deletion.

3. ***Insertions***

   The type of mutation where one or more DNA base pairs are added to a DNA sequence during DNA replication is called insertions.

Q.No.2: What is the Principle of PCR and give its reagents? (5-marks

Answer

Thermo-cycler works on the principle of Peltier effect, which raises and lowers the temperature of the block in a pre-programmed manner. The principle of PCR is based on the use of DNA polymerase which is an in-vitro replication of specific DNA sequences. This method can generate tens of billions of copies of a particular DNA fragment.

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Reagents for PCR process

A complete PCR reaction requires five basic PCR reagents;

- DNA/RNA template
- DNA polymerase
- Primers (forward and reverse)
- dNTPs and PCR buffers.

Q.No.3: What are major types of mutation? (5-marks)

Answer

Several types of mutations can occur in an organism. Following are the major types of mutations;

- **Point Mutation--** Point mutations affect single sites on DNA. It is a single base substitutions which causes sickle-cell disease
- **Missense mutation** – Missense mutations substitute one amino acid for another. Some missense mutations have very large effects. Some have minimal or no effect.
- **Nonsense mutation** – Nonsense mutation is a point mutation where a change in a single DNA base pair results in a stop codon.
- **Insertion mutation** – Insertion mutation is a mutation where one or more DNA base pairs are added to a DNA sequence.
- **Deletion mutation** – Deletion mutation is a mutation where one or more DNA base pairs are removed from a DNA sequence during DNA replication.
- **Duplication mutation** – Duplication mutation is a chromosomal mutation where a section of a chromosome is replicated one or more time.

Q.No.4: Types of polymerase DNA sequence. (5-marks)

Answer

Based on sequence homology, DNA polymerases can be further subdivided into **seven different families; A, B, C, D, X, Y, and RT**.

Some viruses also encode special DNA polymerases, such as Hepatitis B virus DNA polymerase. These may selectively replicate viral DNA through a variety of mechanisms.

On the basis of sequence similarities, DNA polymerases can fall into three groups: **type A, type B and type C**, which have homology to polA (pol I), polB (pol II) and polC (pol III) from Escherichia coli, respectively.

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